of gravitational anomalies—anomalies triggered by physical fields that reside in the bulk.

We are far from sure that the bulk really exists. And it is only an educated guess that, if the bulk does exist, Einstein's laws reign there. And we have no idea whether the bulk, if it exists, contains fields that can generate gravitational anomalies, and if so, whether those anomalies can be harnessed. The anomalies and their harnessing are a rather extreme speculation. But they are a speculation based on science that I and some of my physicist friends are happy to entertain—at least late at night over beer. So they fall within the guidelines I advocated for *Interstellar:* "Speculations . . . will spring from real science, from ideas that at least some 'respectable' scientists regard as possible" (Chapter 1).

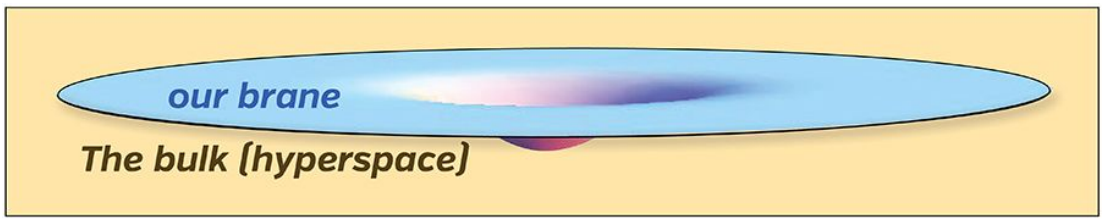


Fig. 3.5. Our universe, in the vicinity of the Sun, depicted as a two-dimensional surface or brane, residing in a three-dimensional bulk. In reality, our brane has three space dimensions and the bulk has four. This figure is explained further in Chapter 4; see especially Figure 4.4.